AGRISENTINEL

Intelligent Agricultural Sentinel for Crop Protection

SMART AI-BASED CROP PROTECTION SYSTEM

AgriSentinel is an AI-powered smart crop protection system designed to help farmers protect their crops from crop-damaging animals while reducing the need for continuous physical monitoring of agricultural fields.

1. Problem

Crop damage caused by animals and wildlife is a major challenge for farmers. Farmers may need to repeatedly visit and monitor their fields, particularly during periods when animals are more likely to enter. This can be difficult, time-consuming, and challenging, especially for farms located far from residential areas. Traditional methods such as physical barriers, human monitoring, or continuously operating deterrent devices may be expensive, labor-intensive, or ineffective when used alone.

2. Our Solution

AgriSentinel combines AI-based animal detection, automated deterrence, live monitoring, incident recording, and alerts into one integrated system. An ESP32-CAM monitors the agricultural field and provides a live video feed. An AI-based detection system analyzes the visual information and identifies specific target animals that may threaten the crops. Instead of activating the deterrence system for every movement, AgriSentinel is designed to respond when a target animal is identified. When a target animal is detected, the system activates an ultrasonic deterrence mechanism using changing frequencies. The aim is to provide a more adaptive deterrent rather than continuously producing one fixed frequency.

3. Smart Monitoring Dashboard

AgriSentinel includes a web-based monitoring dashboard through which the farmer can:

- View the live field camera feed
- Monitor animal detections
- Check detection events
- Review captured images
- View the date and time of incidents
- Observe patterns in animal activity

4. Automatic Incident Recording

Whenever a target animal is detected, AgriSentinel can automatically:

- Capture an image
- Record the date and time

- Store the detection as an incident
- Activate the deterrence system
- Send an alert to the farmer

The collected records can help farmers understand when and how frequently animals enter the field, allowing them to make better decisions about crop protection.

5. Emergency Alerts

When a target animal is detected, AgriSentinel can send an alert to the farmer. This provides immediate awareness of a potential crop threat and allows the farmer to decide whether further action is necessary.

6. Safe and Non-Lethal Approach

AgriSentinel is designed around a non-contact and non-lethal crop protection approach. The objective is not to injure or capture animals. Instead, the system detects a potential threat and activates a deterrent intended to encourage the animal to leave the agricultural area. This supports a more responsible approach to reducing human-wildlife conflict.

7. Where It Can Be Used

AgriSentinel can be adapted for different agricultural environments, including:

- Paddy fields
- Vegetable farms
- Fruit plantations
- Banana farms
- Coconut plantations
- Other crop-growing areas affected by wildlife

For larger farms, multiple monitoring and deterrence units can be deployed to increase field coverage.

8. Future Development

- Solar-powered operation
- Multiple AI camera nodes
- Improved animal classification
- Local data storage and long-term analytics
- Geographical mapping of animal detections
- Additional types of non-lethal deterrence
- More advanced detection-pattern analysis

9. Main Innovation

The key innovation of AgriSentinel is the integration of multiple functions into a single farmer-oriented crop protection system.

AI Animal Identification	→	Threat Verification	→	Changing-Frequency Deterrence	→	Image + Time Logging	→	Farmer Alert

Rather than simply detecting movement, AgriSentinel is designed to distinguish potential animal threats and respond automatically while keeping the farmer informed.

10. Impact

AgriSentinel aims to help reduce crop losses, reduce the need for continuous field monitoring, improve farmers' awareness of animal activity, and encourage safer coexistence between agricultural communities and wildlife.

SMARTER	SAFER	AUTOMATED	FARMER-FOCUSED

THANK YOU

“Smart protection for a smarter agricultural future.”

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